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Land Institutions to Address New Challenges in Africa

Implications for the World Bank's Land Policy

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Abstract

In coming decades, Africa's urban populations will expand, and effects of climate change more keenly felt. In this context, land policies and institutions will be essential to allow urban dwellers to access productive jobs, breathe clean air, and live in decent housing; entrepreneurs, especially women, to leverage land for productive investment; and farmers to diversify, insure against shocks, and accumulate capital. Yet, many African land registries perform poorly, command little trust, and have failed to capitalize on opportunities to improve quality, relevance, and outreach via digital interoperability, use of earth observation, and connectivity. Reform examples and literature suggest that (i) regulatory and institutional reforms are key to building state capacity even in the absence of titling programs; (ii) title issuance in urban areas that aims to improve property tax collection, develop markets for long-term finance, and support infrastructure planning, financing, and construction will have higher returns than rural titling; (iii) issuance of digital georeferenced land use rights can help activate rural factor markets and, if linked to farmer registries, improve subsidy targeting and effectiveness; and (iv) demarcation and transparent decentralized management of public land is essential to attract investment, including climate finance without fueling corruption and manage disputes before they escalate into ethnic violence.

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African countries will face four challenges in the coming decades, all of which can be addressed more effectively if land institutions function well. First, rates of urbanization on the continent are among the highest globally (Henderson and Turner 2020) with urban population expected to triple to more than 1 billion by 2040 (Collier 2017). Urbanization can bring benefits and increase land values through skill-based human capital externalities and productive job creation based on input sharing, labor market pooling, and knowledge exchange (Rosenthal and Strange 2020) but also imposes costs from congestion, contagious disease, and crime (Duranton and Puga 2020). African cities' high land intensity (Esch *et al.* 2023) increases the cost of accessing jobs (Peralta Quiros *et al.* 2019) and cities' carbon footprint (Resch *et al.* 2016).[1] Land and financial markets supported by effective institutions can foster investment in floorspace (Casas-Arce and Saiz 2010) or public services and mobility, the latter preferably financed via property taxes or other forms of land value capture to allow generation of productive jobs in tradable sectors (Venables 2017).

In the past, Africa's agricultural output growth was driven by area expansion (Wollburg *et al.* 2023) but as it pushes into more marginal land, continuation of this trend will be increasingly risky and conflict with environmental objectives. Rural transformation strategies centered on large-scale land-based investment that were pursued by many African countries over the last decade largely failed to increase productivity or generate spillovers due to centralized, nontransparent, and non-competitive public land transfers while input subsidies were ill-targeted and resulted and harmed the environment (Damania *et al.* 2023). Promoting operation of factor markets and structural transformation to reduce high levels of resource misallocation (Restuccia and Rogerson 2017), substitute know-how and capital for labor and land, transfer land use rights for longer periods to foster structural transformation seems a more promising strategy.

Although expected impacts of climate change on mean yields vary by location, climate change will increase yield variability (Conte 2022), often in settings where higher frequency and severity of extreme weather events have already forced farmers to increase defensive spending (Jagnani et al. 2021), reduce migration (Liu *et al.* 2023), or liquidate assets in ways that impair their long-term potential (Aragon *et al.* 2021). Climate shocks also triggered persistent upticks in violent ethnic conflict at the frontier between pastoral and sedentary land use (Eberle *et al.* 2020; McGuirk and Nunn 2020) if land or resource rights were not

[1] Engineering studies that account for cities' total life-cycle GHG emissions suggest 7 to 27 floor buildings are optimal, though high density low rises may be preferable for stable populations (Pomponi *et al.* 2021). African cities have less than 2 floors on average.

clearly documented or local fora for negotiation absent. Without documented land rights, land demands from of renewable or climate finance expansion thus may risk setting off speculative land acquisition (Anti 2021) that may weaken current users' position, or fuel conflict. Many cities' location on coasts or in floodplains increases their exposure to floods or sea level rise. Land markets and the price signals they convey are key to insure against such risks and guide private and public adaptation investments.

Empowering women through asset ownership and control transformed economic activity and governance in today's developed countries (Hazan *et al.* 2019). Land and property account for 40% to 60% of household assets in virtually all countries (OECD 2017) but in most African countries at most a quarter of formal land documents include a woman's name. Gender-sensitive efforts to document land rights increased this to more than 80%, improving women's agency, entrepreneurship, human capital investment, and land and financial market participation. Yet, social norms of inheritance often are persistent as in India (Fernando 2022) or Colombia (Gáfaro and Mantilla 2020) so that even legal changes or documentation of women's rights may not immediately allow them to bequeath land rights (Aldashev *et al.* 2012), reducing productivity (Dillon and Voena 2018). Consistent with the elimination of coverture across states in the US (Geddes and Lueck 2002), customs is likely to change gradually and in response to a combination of factors.

The rest of the paper is structured as follows. The first section discusses benefits from well-functioning registries globally and the channels through which they materialize. It shows that many African land registries fail to meet key preconditions for attaining registry benefits, that traditional institutions can fill the gap only in some settings and argues that such institutional gaps impede operation of land and long-term credit markets. Section two explores the extent to which digital technology opens opportunities to address this and increase affordability, coverage, and quality of land records. It uses examples of reforms to argue that regulatory reform to ensure rights are of high quality and can be accessed and transferred in a way that is affordable was critical to success. Section three discusses policies to fully harness the benefits from documented land rights for urban land value capture, service delivery, planning and expansion; better functioning of rural factor market and links to value chains, and better public land management for climate resilience. Section four concludes.

1. Benefits from land institutions and their institutional preconditions

All over the world, land registries protect land rights and allow transfers at low cost encourage investment and operation of land and financial markets while reducing the need for unproductive spending to defend or fight over rights in addition to providing public benefits. In many African countries, documented rights are affordable only to wealthy men, reducing trust and state capacity, limiting the scope for property tax collection, planning, and land or financial market operation in urban and for better factor market functioning and long-term contracts to foster structural transformation in rural areas. Traditional institutions may fill this vacuum if land transactions remain localized and short term but, unless their operation is regulated, may trigger wasteful forum shopping as land becomes more valuable.

1.1 Global evidence of benefits from well-functioning land institutions

In rural areas, secure property rights reduce expropriation risk, facilitating investment in land improvements (Besley 1995), including via fallowing to restore soil fertility through fallowing (Goldstein and Udry 2008) tree planting (Fenske 2011), or other investments (Lawry *et al.* 2016). Secure tenure increased application of organic manure in rural areas of Niger (Gavian and Fafchamps 1996), Zambia (Dillon and Voena 2018), China (Jacoby *et al.* 2002), Pakistan (Jacoby and Mansuri 2008), and Thailand (Chankrajang 2015) and, if combined with transferability, trees or other improvements (Deininger and Jin 2006). In urban areas, secure rights increased housing investment (Field 2005), aiding human capital development and skill acquisition via health benefits (Vogl 2007) with transfer rights capitalized into land values (Lanjouw and Levy 2002).

The positive effects of documented rights on investment-or prevention of long-term degradation of a resource for short term benefit-are not limited to individual rights. In fact, group rights have advantages if, as for many lands used for grazing, forest, biodiversity, or mineral extraction, gains from individual investment are modest while benefits from resource use accrue at a larger spatial scale (Leonard *et al.* 2021) so that the cost of defining individual rights may exceed incremental benefits (Leonard and Parker 2021). The case for group rights is strengthened if cohesive groups with clearly defined membership and internal as well as external accountability, including graduated sanctions, are in place (Kotchen and Segerson 2020).

For example, legally valid demarcation of indigenous lands reduced deforestation on a large scale in Brazil (Baragwanath and Bayi 2020) and it has been argued that a shift from indigenous to individual tenure (or protected areas that state institutions are ill-equipped to enforce) could result in a net loss of forest cover (Pacheco and Meyer 2022). Recognition of indigenous rights also improved conservation outcomes (Blackman *et al.* 2017), the quality of land management, or income (Peña *et al.* 2017; Vélez *et al.* 2020). In the United States, more than a century after land was assigned to Indian reservation, land of limited productivity had better economic outcomes if it was under tribal ownership than if allotted to individuals with restricted rights (Leonard *et al.* 2020). Clarifying access rights to publicly owned grazing land in the US West similarly increased productivity and use of fallowing as much as privatization (Bühler 2023).

If rights to land are perceived to be contestable, individuals or groups may spend resources on unproductive measures to defend rights (De Meza and Gould 1992) or fight over them (Fetzer and Marden 2017) rather than invest in productive activities. In rural areas, groups with insecure rights (Djezou 2016) often plant trees to establish rights or demarcate borders (Brasselle *et al.* 2002). In Peru's informal settlements, a need to guard property through physical presence that reduced women's participation in labor markets and kids' school attendance was eliminated by awarding formal property rights (Field 2007).

Women's control of household assets a key determinant of bargaining power. Even within the household, land may not be shared optimally due to tenure insecurity, reducing land use efficiency (Udry 1996). Limits on women's ability to bequeath land reduced productivity (Dillon and Voena 2018) and increased fertility (Lambert and Rossi 2016). Building on legal changes to include female co-owners' names on land records made such rights easier to enforce while still allowing flexibility to respect social norms, thereby increasing women's role in intra-household decision-making (Deininger *et al.* 2008; Melesse *et al.* 2018; Muchomba 2017), their bargaining power (Menon *et al.* 2017), or fertility choice (Chakrabarti (2018).

Such measures also increased investment and productivity by women (Ali *et al.* 2014) as well as their participation in land (Holden *et al.* 2011), labor (Field 2007; Goldstein *et al.* 2018), and financial (Ali and Deininger 2023) markets. Yet, norms governing inheritance may be persistent (Fernando 2022) and change only slowly even in response to legal change (Aldashev *et al.* 2012) with rapid change possibly causing backlash (Anderson and Genicot 2015). Legal change (Harari 2019), advice at key life events to increase awareness of women's rights (Field and Vyborny 2019), access to legal aid in case of conflict (Aberra and Chemin 2021), and economic empowerment (Field *et al.* 2021) have all been shown to affect social norms.

Secure documented rights to land as a productive asset provide implicit insurance, increasing individuals' and households' capacity to smooth consumption and take more risky entrepreneurial decisions in Latin America (Aragón *et al.* 2020; Di Tella *et al.* 2007) and Europe (Sodini *et al.* 2021). In China, assignment of rights over urban flats to women contributed to female economic empowerment (Wang 2014). Pilots to register all rural land, together with measures to ease transferability, almost doubled the likelihood of nonfarm enterprise startup (Deininger *et al.* 2019) while increasing land market transfers to (young) producers, off-farm labor supply by older farmers, and profits through a shift from grains to vegetables and other higher value crops (Deininger *et al.* 2020). Full roll-out of registration increased entrepreneurship by professionals and larger farmers and encouraged more urban residents to start new businesses in rural areas due to better land and labor market functioning (Bu and Liao 2022).

Asymmetric information and risk lead to credit rationing that can be reduced through the use of collateral (Stiglitz and Weiss 1981). As land is immobile and difficult to destroy, it is ideal collateral (de Soto 2000). China's privatization of state housing alleviated credit constraints by allowing individuals to capitalize on the value of their real estate (Wang 2012). In France, exogenous shifts in individuals' access to collateral persistently increased their propensity to assume debt, take up entrepreneurship, and establish larger firms (Schmalz *et al.* 2017). Legal changes that affected the use of land as collateral increased property values directly in the US (Zevelev 2021) with strong collateral effects also found in Australia (Atalay and Edwards

2022) and Singapore (Agarwal and Qian 2017). By contrast, illiquid rural land markets (Lawry *et al.* 2017; Stein *et al.* 2016), obstacles to enforcing collateral in case of default, or unaffordable cost of registration (Galiani and Schargrodsky 2016) often disappointed hopes of titling to relive credit constraints quickly or limited collateral effects to larger borrowers (Carter and Olinto 2003; Zhang *et al.* 2020).

1.2 Public benefits from effective land institutions

Land registries also generate public benefits. Beyond land prices helping to improve efficiency of resource allocation, documented land rights allow to (i) regulate land use to manage externalities; (ii) foster public investment through land value capture; and (iii) better manage public land. Mandatory publication of transaction prices can reduce gaps in access to information (Seifert *et al.* 2021) and reduced price dispersion in Israel (Ben-Shahar and Golan 2019). Easier property appraisal increased loan-to-value ratios and mortgage access (Jiang and Zang 2022), implying that publicity of land prices can reduce financial market transaction costs by making it less costly to comply with macro-prudential regulations. Land registries also provide robust data on individual wealth and,[2] joint with registries of beneficial company ownership (Neef *et al.* 2022), are key to monitor cross-border real estate investment (Alstadsæter and Økland 2022) and its use for money laundering (Jeanne and Le Guern Herry 2022).

Public goods such as sewers (Coury et al. 2021), broadband access (Ahlfeldt *et al.* 2017), road quality (Gonzalez-Navarro and Quintana-Domeque 2016) and location-specific amenities including trees (Han *et al.* 2021) or coastal preservation (Severen and Plantinga 2018) are capitalized in land values. The same is true for dis-amenities such as exposure to air pollution (Chang and Li 2021), radon (Pinchbeck *et al.* 2020), nuclear plants (Coulomb and Zylberberg 2021), or earthquake risk (Singh 2019). Land prices can thus help assess the cost of policies such as zoning regulations (Hilber and Vermeulen 2016), historical preservation mandates (Hilber *et al.* 2019), congestion pricing (Tang 2021), or limits on land market participation (Hartley *et al.* 2023; Lawley 2018).

As land is immobile, taxing it will generate fewer distortions than taxes on more mobile factors such as labor or capital. Recurrent land taxes as well as surcharges to capture appreciation of land values due to public investment are ideally suited to finance local public good provision. They have been shown to increase accountability in African contexts (Weigel 2020) and strengthen the social contract (Besley *et al.* 2013). In light of secular land appreciation (Knoll *et al.* 2017) that is further enhanced by public investment in mobility (Gupta *et al.* 2022), land value capture and use of future land tax receipts as collateral for bonds often provide vast unrealized opportunities.

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[2] Data on property transactions, possibly linked to other administrative records, can overcome shortcomings of household surveys such as high nonresponse rates by the very wealthy (Brummet *et al.* 2018) so that consumption (Eika *et al.* 2020) or wealth (Fagereng *et al.* 2020) can be measured more reliably, allowing to document wealth inequality by gender (Frémeaux and Leturcq 2020).

Delays in land acquisition or permitting pose bottlenecks to infrastructure construction and expansion of renewables globally. Clear land records and publicity of land prices to allow transparent valuation can help discern projects with large public benefits from mediocre ones and prevent the former from being derailed by holdouts (Schafer and Singh 2018), political interference (Ghatak *et al.* 2013), or litigation (Gandhi *et al.* 2021). It can also make it easier to implement mechanisms such as land sharing (Lozano-Gracia *et al.* 2013), mixed income development (Blanco and Neri 2022) or in kind compensation (Ghatak and Ghosh 2011) to help households who lose land to such investment but whose capacity to capitalize on cash-only compensation may be limited (Mezgebo and Porter 2020; Sha 2022) to maintain their living standards.

1.3 Institutional preconditions and actual performance

Realizing the benefits from clear land rights requires public land registries to make reliable, complete, and enforceable information on rights and encumbrances (mortgages, court cases, pending land acquisition etc.) pertaining to a given land parcel available to interested parties. Such institutions, the establishment of which often required decades or centuries in today's developed countries (d'Arcy *et al.* 2021), comprise a spatial and a textual component. The former, referred to as *cadaster*, describes the boundaries of the objects to which rights are defined while the latter, called the *register*, documents the type of rights held to land and allows them to be ranked by priority. Making information on objects, subjects, and the type of rights public and ensuring enforcement by state organs if needed reduces transaction costs and increases competition in land and financial markets as buyers need to spend less resources on verifying the authenticity of rights they are about to acquire. This expands the range of potential market participants, the complexity of rights that can be held or transacted, and the length of time for which contracts can entered and enforced.

In many African countries, gaps in quality, coverage, or publicity of registered land rights increase the cost of acquiring, transferring, or enforcing such rights. Household survey data in table 1 show that registration of land rights remains limited and biased by wealth and gender: levels of informality exceed 70 percent in urban and 90 percent in rural areas in all countries with data. Households in the top quartile of the urban expenditure distribution are between 2 and 10 times more likely to have registered land rights than those in the bottom and more than 70 percent of registered rights are held exclusively by men.

A main reason for such wealth or gender bias is the cost of title acquisition, reported to be above US$1,000 per parcel in rural Madagascar (Jacoby and Minten 2007) or Ghana (Agyei-Holmes et al. 2020), and close to US$500 in Benin (Mekking *et al.* 2021) and Zambia (Honig 2021). Regulations that require a high-precision survey and physical boundaries monumentation to support titles are a key factor that increases costs. However, a survey just describes the object to which rights are held without increasing security of rights or the ability to enforce them (Arruñada 2018). Requiring it may, via lock-in effects, reduce rather than increase land transfers–as was the case over long periods in the United States (Allen and Leonard

2021). Moreover, as number of licensed surveyors in many developing is limited, increases in demand will often only increase scarcity rents, as apparently happened in Tanzania (Manara and Regan 2021).[3]

Regulatory change to reduce survey standards is thus an important first step to make land documents in Africa more affordable. The UK which has, for over a century, used topographical maps rather than high precision surveys to maintain a title registry, shows this is possible. In Africa, the example of Rwanda which adopted regulations allowing the use of high-resolution satellite imagery (under a 'general boundary' concept), to then quickly issue titles to 11 million parcels at a unit cost of less than US$6 (Nkurunziza 2015) highlights the scope for drastically reduce affordability of title, especially in urban areas. This suggests that delivering urban title should cost at most US10-20, in line with urban residents' willingness to pay for title. Evidence from the Democratic Republic of Congo (Balan et al. 2020), Tanzania (Ali *et al.* 2016; Manara and Regan 2021),[4] Zambia (Ali *et al.* 2020a), and Ghana (Ehwi *et al.* 2021) shows that informal residents are willing to spend significant resources to have their land rights properly documented. Reducing cost of service delivery and passing gains to customers would thus allow to achieve almost complete coverage with proper title—rather than inferior substitutes for which there is no demand[5]—at close to full cost recovery.

Issuing titles will not be sustainable if registration of subsequent transfers is unaffordable. Three factors that commonly increase registration costs are (i) transaction taxes, (ii) inefficient service delivery often due to reliance on complex paper-based processes and a desire to recover at least some of registries' operational cost;[6] or (iii) using registration fees to generate revenue by adding unrelated charges (Ali *et al.* 2016), including rents for notaries (Gutierrez and Molina 2020).[7] High fees contributed to undermining registry sustainability in Africa (Atwood 1990; Pinckney and Kimuyu 1994), and caused a returns to informality (Galiani and Schargrodsky 2016) after titling programs almost everywhere.[8] As use of land as collateral requires mortgages to be registered, high fees will undermine credit market effects even if titling may have positive effects otherwise (Agyei-Holmes *et al.* 2020; Fabbri and Dari-Mattiacci 2021).

Column 1 of table 2 shows that high transfer taxes drive up registration cost to an average of more than 7% of property in Africa while columns 4 to 6 highlight the scope for reducing cost of service delivery via steps such as fully digitizing registry and cadaster and ensuring interoperability between them, that could also

[3] High cost led owners to abandon titling processes halfway (Ali and Deininger 2022a), resort to informal substitutes (Chimhowu and Woodhouse 2005) such as sales contracts (Lavigne Delville 2002) or tax receipts (Goodfellow and Owen 2020), and biased access to documents in favor of the wealthy who were often less able to make productive use of land (Sitko and Jayne 2014).

[4] In Dar es Salaam, residents of low-income unplanned settlements participated in a saving scheme to pay an average US$35 for title (Ali *et al.* 2016) and use of private information shows a willingness to pay of up to US$200 (Manara and Regan 2021).

[5] Tanzania's experience with 'residential licenses', a non-transferable temporary document certifying use rights issued without proper checks (sometimes to tenants posing as owners), implying that in some cases banks were unable to foreclose on land pledged as security for loans using such documents (Kusiluka and Chiwambo 2019) highlights that low cost should not be equated with inferior rights or quality; in fact there was virtually no demand for these documents even from low income neighborhoods (Manara 2022).

[6] A detailed discussion of strategies is beyond the scope of this paper. See (Ali *et al.* 2021) for more detail.

[7] In Peru, notary pressure led to closure of a well-functioning low-cost registry, increasing informality (Gutierrez and Molina 2020).

[8] In urban Argentina, registering inter- or intra-family transactions costs 27% and 20% of property value respectively (Galiani and Schargrodsky 2016). High cost triggered 'de-formalization' also in Jamaica (Barnes and Griffith-Charles 2007) and the Philippines (Maurer and Iyer 2008).

increase registry quality. High cost reduces coverage with less than one in eight countries having most land parcels in the capital registered or mapped (col. 5 and 6). Together with reliance on paper documents reduces the scope for enforcement; only 4 of 53 African countries are courts able to resolve a 'typical' land conflict in first instance in less than a year.

Perception data from 34 African countries also suggest that less than 20 percent of respondents are confident they will be able to obtain reliable information on land rights from the registry and almost two thirds (61 percent) doubt the impartiality of land registries and they believe rich people will be able to fraudulently register land that does not belong to them (see table 3). This points towards vast scope to prevent a vicious cycle of eroding trust in state institutions by moving to a fully digital system, restructuring workflows, and reduction of registration costs at least in urban areas. This can be done using off the shelf technology but may be opposed by those benefiting from petty corruption, implying that a political decision is needed.

1.4 Opportunities and limitations of traditional institutions

In rural areas, customary institutions that, through chiefs, long managed use of common land such as pasture and forest for joint benefit and allocated arable land for cultivation under secure long-term rights, may be able to may fill the trust deficit discussed above and administer land in a more cost-effective and flexible way than would be possible for formal institutions (Boone 2017). Studies show that, if chiefs are bound by informal accountability channels (Baldwin 2018), local presence provides them with access to private information (Basurto et al. 2020) that allows resolving local disputes quickly, fairly, and without demanding bribes (Winters and Conroy-Krutz 2021), and providing public goods (Tourek et al. 2020).

Support for customary institutions is rooted in the fact that they provide lineage members with an innate right to land use rights with restricted transferability (Andolfatto 2002) that acts as a safety net (Baland and Francois 2005), allowing to cope with idiosyncratic shocks (Beck *et al.* 2019), transfer land locally, and allow expansion by efficient land-poor farmers (Deininger *et al.* 2017). Evidence from Uganda (Baland *et al.* 2003), Kenya (Jin and Jayne 2013), Ethiopia (Teklu and Lemi 2004), Zambia and Malawi (Chamberlin and Ricker-Gilbert 2016) shows productivity benefits from land rental. Welfare effects are usually positive as rents received by owners exceed their potential return from self-cultivation (Deininger *et al.* 2011) with women often benefiting disproportionately (Bezabih *et al.* 2016; Wineman and Liverpool-Tasie 2017).

Survey data in table 4 show that traditional leaders continue to play an important role by resolving land disputes (49 percent), governing the community (36 percent), and allocating land (31 percent) with 55 percent of respondents wanting traditional leaders' influence to increase. Recognition of the advantages of traditional authorities (Bruce and Migot-Adholla 1994) led to a wave of laws to recognize customary tenure across Africa in the early 2000s (Alden Wily 2018b), though there has been very little progress on passage of subsidiary regulation to guide their implementation.[9] As long as land remained relatively abundant, this meant laws were ignored (Bubb 2013). However, as land values and demand for land by outsiders increases, the complexity of decisions and the stakes involved will increase, implying that regulation, especially in three areas, can benefit all parties:

Accountability and responsibilities: Colonial intervention (Frankema and van Waijenburg 2013), land concessions (Lowes and Montero 2020), or co-optation (de Kadt and Larreguy 2018) often weakened traditional leaders' downward accountability, reducing locals' voice (Acemoglu *et al.* 2014), governance (Ali *et al.* 2020c), and resource management (Mihaylova 2023) across and within countries (Ali *et al.* 2019c). As land values increase, this may lead chiefs to behave like owners rather than custodians of land (Lanz *et al.* 2018) by transferring land to wealthy outsiders for their own benefit (Chimhowu 2019) even if such transfers reduce social welfare (Leeson and Harris 2018) and are difficult to enforce (Lentz 2010).

Recognition of customary tenure that is not accompanied by regulations to define chiefs' area of influence, their responsibilities, and ways to enforce these may thus create parallel systems and reinforce wealth bias as the affluent may resort to 'forum shopping', i.e. purse land claims with formal and customary institutions (Eck 2014) in parallel. In fact, formal recognition of customary tenure could make rights more contestable and access to them more unequal, as in Uganda (Gochberg 2021) or Zambia (Honig 2017) if some chiefs start to charge for services they had earlier provided for free (Honig 2021), even if these are financed by donors (Adam *et al.* 2021; Green and Norberg 2018). A RCT in Zambia indeed shows that issuing use right certificates without clarifying how these relate to statutory titles or will be maintained has no economic impact (Huntington and Shenoy 2021).

Contract enforcement with outsiders. In West Africa, migrants received "perpetual" rights to incentivize cash crop investment when land was abundant but saw agreements re-interpreted or revoked (Woods 2003) with greater land scarcity (Bambio and Bouayad Agha 2018). 'Outsiders' are disproportionately at risk of land conflict as in Burkina Faso (Linkow 2016), Côte d'Ivoire (Fenske 2010), and Uganda (Mwesigye and Matsumoto 2016). This triggered defensive but unproductive investment (Bros *et al.* 2019), reduced productivity of land use, and triggered violent conflict in Burundi, Rwanda, the Democratic Republic of Congo, northern Uganda, southwestern Côte d'Ivoire, and Senegal (Boone 2017).

Traditionally, as land became more scarce or as transactions with outsiders created opportunities to access technology or markets (Fenske 2014), customary systems tended to gradually individualize (Guirkinger and Platteau 2014), a transition that could be accelerated if other options to insure or manage externalities became

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[9] The importance of customary institutions is also recognized by the "Framework and Guidelines for Land Policy" adopted by African Heads of State in 2010 (https://www.un.org/en/land-natural-resources-conflict/pdfs/35-EN-%20Land%20Policy%20Report_ENG%20181010pdf.pdf) and the FAO Voluntary Guidelines on responsible governance of tenure (https://www.fao.org/3/i2801e/i2801e.pdf) .

available (Delpierre *et al.* 2019). As tenants, especially outsiders, will make 'optimum' levels of investment in soil fertility (Ricker-Gilbert *et al.* 2022) or irrigation and drainage (Fan and Yeh 2019) only if long-term contracts can be enforced, regulation is needed to define what type of contracts can be entered in with group members as well as outsiders and how such contracts are to be documented and enforced.

Managing local resources and resolving conflict: Sedentary and pastoral groups have long engaged in a mutually beneficial symbiotic relationship that ensured sustainability of the latter via access to floodplains or wetlands in the dry season. Unforeseen droughts or unilateral land use intensification can transform this relationship from a complementary to a confrontational one especially if fora for conflict resolution are one-sided (Bergius *et al.* 2020) or rights to critical resources (e.g. watering holes or migration routes) are undocumented. Data for almost three decades in Africa show that, by exacerbating competition over scarce wetlands, droughts in pastoral areas triggered agro-pastoral conflict in neighboring agricultural areas (McGuirk and Nunn 2020). Political participation and access to fora for land dispute resolution reduced this effect (Eberle, Rohner and Thoenig 2020).

Beyond this specific case, greater land scarcity and more frequent agricultural productivity shocks alter the opportunity cost of fighting relative to production (McGuirk and Burke 2020). Without negotiation, this can increase land conflict (Berman *et al.* 2021) in a way that may affect neighboring areas (Harari and La Ferrara 2018) and persist or escalate over time (Moscona *et al.* 2020). ACLED data that allow identifying conflicts related to land or resources in table 5 reports the share of violent conflicts related to land and natural resources for three periods from 2007 to 2022, show that frequency of such violence more than doubled overall and quadrupled in West and Central Africa.

2. New opportunities and examples of reform

Digital interoperability, connectivity, and use of remotely sensed data allow to reduce the cost of registry establishment and maintenance while increasing quality of land records, the uses to which they can be put, and the ease with which right holders can update or exchange rights. Reforms in select countries and global experience illustrate the benefits from harnessing this potential if the regulatory framework is appropriate but also the fact that, in the absence of adequate regulations, digital tools can create parallel systems that may exacerbate pre-existing inequality or increase conflict.

2.1 The potential of digital technologies

Advances in cloud computing and digital interoperability, use of remote sensing, and connectivity create opportunities to reduce the cost of establishing and running registries improve the quality and publicity of land records and in doing so establish such records as a foundational layer for a national spatial data infrastructure to complement benefits from digital IDs (Hanedar *et al.* 2023) for transparent implementation of public programs public programs (Haseeb and Vyborny 2022; Muralidharan *et al.* 2016).

Digitization and interoperability can reduce the cost and of establishing, maintaining, and accessing land registries while increasing publicity and transparency by establishing clear workflows. Even without changes in record quality, easier record access through digitization of textual records increased mortgage registration and urban credit access by more than 10 percentage points in India (Deininger and Goyal 2012) and reduced market frictions, allowing land transfers to more productive farmers that improved agricultural productivity in Pakistan (Beg 2022). Unambiguous identification of land parcels and automatic digital links with the civil registry or with registries for mortgages, court decisions, legal entities or building permits can improve quality and reduce cost of maintaining records by ensuring information is consistent or by automatically requiring action if, for example, the registered landowner's civil status changes.

Digitization of spatial data adds to the above in two respects. First, in many countries, documents conveying rights to land issued by different entities in the past may have created overlapping rights.[10] Digital overlays of the maps or sketches substantiating such land grants is the fastest way to identify competing claims and, if they exist, identify was to legally extinguish dormant or unused rights. Second, land use classification based on remote sensing can, combined with digital boundaries, help ascertain if or how land is used in near real time. Even if public land has never been mapped, having boundaries of private land documented will make it easy to identify land that is cultivated but not privately owned as possible encroachment and take follow-up action accordingly to ensure public land is managed in ways that benefit local communities.

For example, in Malawi, 1.35 million ha or 25 percent of the country's arable area, were leased to estates via 30-year contracts in the late 1980s in blocks of some 30 ha each. Digitization of associated maps revealed that some 140,000 hectares had been allocated twice and 28% of estates had areas overlap with other estates while overlays with land use maps based on satellite imagery allowed to identify unused estate land (Deininger and Xia 2018). Combining spatial with textual data suggested that charging competitive lease rates for all estate land could generate recurrent revenue equivalent to 5% of public spending. Billing all estates and using non-payment a reason to initiate legal proceedings to cancel estates' legacy rights to land could help clarify tenure and free up land to be reassigned to socially more desirable uses.

Remote sensing: High resolution satellite or drone imagery reduces the cost of documenting land rights by allowing easier identification of property boundaries for local confirmation in a participatory adjudication process. In urban areas, optical data allows to estimate city compactness (Burchfield *et al.* 2006) and, together with radar and other data, provide estimates of floorspace at national (Arribas-Bel *et al.* 2021) or

[10] Brazil is an extreme case where the total area with documented rights is about 5 times the size of the national territory (Stabile et al. 2020).

global scale (Esch *et al.* 2022). Together with maps of environmental risk, this allows analyzing the extent to which property prices reflect such risks (Hino and Burke 2021).

In rural areas, free remotely sensed imagery dating back to the late 1970s from the Landsat archive or since 2017 from the Copernicus Sentinel constellation (Tassa 2020) on the Google Earth Engine (GEE) cloud computing platform (Gorelick et al. 2017) also allows to monitor land use changes over long time periods at scale (Burke et al. 2020) to, for example, establish land use benchmarks.[11] Combined with ground truth data (Burke and Lobell 2017), this allows to obtain data on land cover (Lobell et al. 2020) or carbon stocks (Csillik et al. 2019; Harris et al. 2021) as a basis for climate finance.[12] Such data, together with information form public registries can be used to support industry commitments to 'zero deforestation' (Moffette *et al.* 2021) in soy (Heilmayr *et al.* 2020) or oil palm (Lee et al. 2020) value chains.

Brazil's DETER system that monitors changes in forest cover in near real time on the GEE platform and triggers follow-up action in high risk cases (Doblas et al. 2022) illustrates the scope for property boundaries to monitor land use. The system was instrumental to reduce deforestation (Braganca and Dahis 2021) by making access to subsidized public credit contingent on compliance with regulations requiring to keep a share of land under forest (Assunção et al. 2020). The information was also used to 'blacklist' municipalities with high levels of deforestation (Assunção and Rocha 2019), reducing deforestation by 40 percent, with potentially even bigger effects had selection focused on high risk municipalities (Assuncao *et al.* 2019).

Connectivity: The recent expansion of mobile phone ownership and coverage provides additional benefits in three respects. First, it allows enforcement of joint rights by requiring digital consent from all co-owners before allowing any transaction to be registered–and depositing one-time or recurrent payments associated with it automatically in co-owners account via mobile money. This can eliminate transfers of jointly owned land by husbands without knowledge by female co-owners, a practice that anecdotal evidence suggests was widespread (Meinzen-Dick *et al.* 2019) but, if the large and persistent effects of direct wage deposit on social norms reported in the literature (Field *et al.* 2021) are a guide, help empower women economically.

Connectivity reduces the transaction cost of market participation, increasing competition and expanding the range of potential contractual partners in land markets (Hüttel *et al.* 2013), especially if price data is also publicized (Eerola and Lyytikainen 2015). Digital connectivity allows contracts to be conditional on specific outcomes, e.g. weather at granular scale for index insurance (Masiza *et al.* 2022); inter-linked to

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[11] Landsat offers a 30-meter pixel size and 16-day revisit frequency with a pixel size of 10 meters and a 5-day revisit frequency. Standard products are increasingly made available for free, including for agricultural monitoring in Africa (Michler *et al.* 2022; Nakalembe *et al.* 2021).
For example, time series data show that between 2003 and 2019, cropped area increased by 9 percent to a total of 1,244 million ha. The annual rate of expansion almost doubled, mostly in Africa, where 49 percent of expansion replaced natural vegetation and tree cover (Potapov et al. 2022). Global soybean area more than doubled from 26.4 million to 55.1 million ha over the same period (Song et al. 2021).

[12] Machine-learning based approaches to image segmentation may expand on this by allowing automatic generation of boundaries for millions of agricultural fields at a country level as a potential basis for managing use rights.

protect against downside risk (Carter 2022), e.g. by bundling high yielding seeds with insurance (Bulte *et al.* 2020) in a way that could also be used to replace sharecropping with cash rental; or more complex and conditional on behavior by other market participants (Bryan *et al.* 2022).

By increasing competition in electronic auctions, connectivity can also increase benefits from public land as in Ukraine where a shift to electronic auctions by local governments immediately increased prices by 175 percent (Deininger *et al.* 2023). Evidence from China, where non-transparent transfers of public land have been a key source of corruption (Chen and Kung 2019) further suggests more transparent transfers led to better performance in terms of job creation and investment (Kahn *et al.* 2021).[13]

2.2 Examples of institutional and regulatory reform

Rwanda is the only African country with a fully digital nationwide registry and cadaster. After the 1994 genocide that partly originated in contests of land (Andre and Platteau 1998), the country completely revised its legal framework, conducted a carefully evaluated pilot, and ensured that regulations could be drafted and passed quickly to respond to issues as they emerged (Ngoga 2018). This allowed to issue title to the country's 11.5 million parcels in 2011–2013, with clear benefits: for 86 percent of land parcels, ownership documents included a woman (Ali *et al.* 2017), and soil conservation investment (Ali, Deininger and Goldstein 2014) and activity in land and mortgage markets (Ali *et al.* 2015) increased markedly. The move to a flat fee radically reduced cost of registration and moved the country from rank 137 to the top of the 'Registering Property' part of the World Bank's 'Doing Business' ranking between 2005 and 2015.[14]

The program also improved state capacity: incremental urban property tax revenue from properties that were added to the registry under the program would allow to recoup cost in less than a decade or three years with rates unchanged or uniformly set at 1 percent of property value, respectively. Having established a digital registry, interoperability with the National ID, the mortgage registry, and systems for valuation, building permitting, and land use control was established to enhance land record quality. The ability to obtain information on rights to each land parcel via SMS message enhances publicity and efforts to move towards fully paperless titling are currently underway (Akumuntu and Potel 2022).

In Lesotho, women had very few rights until a 2006 law allowed them to own property independently. Two Land Acts passed in 2010 then provided the basis for institutional and regulatory reform to, among others, link registry and cadaster, simplify workflows, reduce time required for registration as we;; as its cost, and strengthen women's rights via a presumption of joint ownership for property acquired in marriage under

[13] Anonymized cell phone call records have also been used to assess mobility and commuting flows and to infer the spatial distribution of income at workplace and residence by skill group (Kreindler and Miyauchi 2021).

[14] The flat registration fees implied that, registration cost, relative to property value, were disproportionately high for rural land. A survey showed this led indeed to informality in rural areas and, by collecting evidence on willingness to pay, helped identify options to adjust fee structures to help deal with this without jeopardizing the registry's ability to sustain itself from fee revenue (Ali *et al.* 2021).

community of property. This regulatory reform was then followed by an effort to systematically register land rights in urban informal settlements.

Analysis of these measures based on two decades (2000-2019) of registry data allows to draw three lessons. First, in the short term (2007-14), policy reform that reduced registration cost and improved record quality facilitated large increases (9 and 4-fold, respectively) in land and mortgage market participation, largely by those who already had registered rights or took advantage of lower cost due to reform. In the longer term (2000-2019), both policy reform and titling increased access to credit markets including for female (co-) owners. Effects of policy reform were much larger than those of titling, partly due to differences in parcel characteristics. Interestingly, legal reforms to empower women affected outcomes only once legal reforms and systematic titling had made access to documented rights more affordable, suggesting that inefficient service delivery can prevent gender sensitive legislation from having impact (Ali and Deininger 2023).

The case of Nigeria illustrates how, in a situation where regulatory reform is difficult, satellite imagery can be used providing a basis for property taxation to increase public revenue and create demand for better documentation of land rights. Building on the example of Lagos state that established a comprehensive property tax map and simplified collection of various property-related fees into a single land use charge starting in 2010, a World Bank-supported 'program for results' provided technical assistance, in the form of high-resolution imagery and three-dimensional vector data covering all urban structures; software for field data collection; and regular peer learning. The goal was to enable states to collect data on owners, parcel, and structure characteristics for at least 50% of urban structures to receive a lump-sum payment. In about 12 months, 34 of 36 states managed to reach the target, collecting information for more than 8 million properties that is now used for to plan and improve public service delivery and collect property taxes, an action that has already given rise to demands for documented land rights.

In a rural setting, the case of Mexico shows how a regulatory framework to document individual as well as community rights can implemented in a decentralized and sustainable way to show impact in several dimensions. The country documented customary rights by groups (*ejidos*) to more than 100 million hectares in the early 2000s in three steps.[15] First, *ejidos* were registered as legal entities with clearly defined statues and organs of self-governance (the executive, assembly, and an oversight committee) to streamline internal governance and allow entering enforceable contracts with outsiders as a group based on assembly decisions. Second, external borders of the *ejido's* land holdings were drawn, community and individual agricultural or housing parcels were demarcated locally and registered in the central National Agrarian Registry (*Registro Agrario Nacional* (RAN)). Third, all decisions were ratified by the assembly, supervised by a

[15] By 2008, the voluntary Program for the Certification of Ejido Rights and Titling of House Plots (*Programa de Certificación de Derechos Ejidales y Titulación de Solares* (PROCEDE)) had certified 98 percent of *ejidos* with more than 100 million hectares based on communities' request.

network of independent para-legals from the publicly supported Land Attorney's office (*Procuradoria Agraria*), with enforcement through a dedicated set of courts in each state.

A procedure for *ejidos* to exit the customary regime and move to the private system by individualizing land rights was also put in place, together with safeguards, in particular a requirement for such decisions to be approved by 75 percent of members in a centrally supervised vote.[16] In practice, less than 10 percent of *ejidos*, mostly those closer to cities and with higher incomes, levels of literacy, and reliance on industry and services exercised this right (Ramírez-Álvarez 2019). Although under the customary regime, individual land cannot be transferred to outsiders or used as collateral, clear assignment of rights through the reforms provided large economic benefits as the program facilitated mechanization and increased productivity (de Janvry et al. 2015); fostered structural transformation via migration (Valsecchi 2014) and investment (Dower and Pfutze 2013); changed local political dynamics (de Janvry *et al.* 2014); and reduced violence by constraining authorities' discretion in reallocating land rights (Castañeda Dower and Pfutze 2020).

In the African context, the scope for scaling up decentralized use rights issuance quickly, equitably, and effectively is illustrated by the case of Ethiopia where, in 2003–05, land use certificates were issued to more than 18 million parcels by 6 million households at low cost (< US$1 per parcel) in an equitable way led by elected councils (Deininger *et al.* 2008). The program facilitated climate change adaptation (Rampa and Lovo 2023) and land-related investment (Holden *et al.* 2009; Melesse and Bulte 2015), reduced land conflict (Di Falco *et al.* 2020), and strengthened women's rights (Kumar and Quisumbing 2015) and their bargaining power (Melesse, Dabissa and Bulte 2018). It also increased participation in local rental markets (Holden, Deininger and Ghebru 2011), with benefits that exceeded the cost (Deininger *et al.* 2011). Although land sales remain prohibited and even rentals restricted in several regions, certification realized about a fourth of the potential productivity gains from eliminating factor misallocation (Chen *et al.* 2022).

Without guidance on updating, villages improvised record maintenance, leading to highly variable record quality (Cochrane and Hadis 2019). To unify procedures and prevent a reversion to informality, a program of 'second round' certification created a computerized and fully georeferenced registry with more than 22 million records of public and private land at US$6 per parcel (Zein 2021). Launched in 2020, this system now operates in some 350 *woreda* (county) offices in 6 regions, where it is used to register 24 transaction types using fully digital auditable workflows that incorporate and enforce regional differences in regulations. More than 750,000 have thus far been registered. Although limited network capacity precluded connecting directly to users' or service providers' mobile phones (for example to allow digital initiation of transactions or lodging of surveys), plans include linking the land registry to the national ID system and

[16] This contrasts with "on-demand" opportunities for exit by individuals via private titling that is practiced in many African countries, that threaten to hollow out the system from within and may give rise to inequality and rent seeking.

using it as a basis for farmer-specific agricultural advice, public land use monitoring, and expansion of coverage into urban areas.

2.3 Implementation challenges

In Benin, rural land use rights in 300 randomly selected villages were, in 2008-11, documented via low-cost demarcation of village boundaries, followed by elaboration of rural land use plans and demarcation of individual parcels. Village level interventions reduced tree cover loss, fires, land clearing, and inter-village border conflicts while increasing trust in local institutions and local participation in managing communal land and resources (Wren-Lewis *et al.* 2020). Individual parcel demarcation increased and investment in tree planting and fallowing, especially on parcels managed by women (Goldstein *et al.* 2018).

As rights to land demarcated in this way remained conditional on conversion to full title via a costly and often discretionary process within few years (Lavigne Delville 2019),[17] the program's long-term effect was more ambiguous: it increased land conflict (Arruñada *et al.* (2022) and had no impact on income, education, credit access, migration, or altruism (Fabbri and Dari-Mattiacci 2021). Regional heterogeneity whereby wealth bias in access to secure rights introduced by the project reduced support for 'modern' land institutions in marginal locations (Fabbri 2021) while increasing support for the rule of law in more affluent ones, suggests a regionally differentiated approach that formalizes existing arrangements gradually, with costs commensurate to the incremental benefits, may be more sustainable and affordable.

Tanzania's 1999 Village Land Act, hailed as one of the most progressive laws in Africa at the time of its passage (Alden-Wily 2003), shows legal provisions to recognize use rights can be rendered ineffective by a combination of (i) costly and complex procedures that favor the wealthy and create space for discretion by officials; (ii) failure to publicize land rights and restrictions; and (iii) lack of clarity on enforcement. Elaborate technical requirements laid down in regulations governing the preparation of village land use plans as a first step towards issuance of a Certificate of Village Land (CVL) imply that two decades after passage of the Act, less than 10 percent of villages had such plans (Huggins 2018). Despite the complex process for elaborating them, land use plans' lack of publicity impedes enforcement and several are reported to have been changed in ways that reduce pastoralists' rights without their consent (Bergius *et al.* 2020).

Cases of government refusing to enforce registered village boundaries as land was 'unused' reduced locals' as well as investors' confidence in the ability to enforce rights, prompting several agri-business ventures to cancel their plans (Engström and Hajdu 2018). The regulations governing issuance of use rights on village land mandate a process that is virtually undistinguishable from that for issuing formal title, with costs of

[17] When it closed in 2013, the program had issued only about 5,000 of the expected 70,000 titles issued. Certificates issued by the project were abolished in 2017 and replaced with a similar but 7-15 times more expensive document (Lavigne Delville 2019). In Côte d'Ivoire, the need for title increased insecurity by tenants as they allow landlords to opportunistically renege on informal agreements that were now 'illegal' (Colin 2013)

several hundred dollars (Stein *et al.* 2016), thus replicating the wealth bias that motivated adoption of a separate Act to govern village land to start with. Moreover, issuance of such documents to individual parcels is on demand rather than comprehensive for all villagers, threatening to increase intra-village inequality and undermine the scope for collective action in managing village land (Mramba 2023).

Lack of implementing regulations for progressive laws led to similar challenges in neighboring countries: In Mozambique, efforts to document community rights, amply supported by donors, had limited impact (Quan *et al.* 2013). In Uganda, regulations to govern issuance of 'low-cost' use rights under the 1998 Land Act created a level of complexity and cost—largely salaries—that cannot be justified economically (Hunt 2004). In Zambia, lack of legal provisions to accept electronic signatures have held up private sector efforts to expand the supply of land documents for years (Sagashya and Tembo 2022).

In Kenya, implementation of far-reaching changes to decentralize land administration written in the 2010 Constitution was delayed (D'Arcy and Nistotskaya 2019) partly due to limited information sharing (Boone *et al.* 2019) and use of land for patronage (Hassan and Klaus 2020). Implementation the 2016 Community Land Act, a legislation passed with high expectations (Alden Wily 2018a), is held up by a lack of regulatory clarity (Odote *et al.* 2021). Computerization is argued to have the potential of triggering a shift in access to rights through across local governments (Dyzenhaus 2021) but its long-term impact will depend on the legal validity and enforceability of digitized documents being properly regulated.

Malawi's 2016 Land Law puts all customary land under traditional land management authorities (TLMAs) and mandates demarcation of TLMAs' land as a first step toward issuing low-cost use right certificates. While the legislation was highly contested and much delayed (Chikaya-Banda and Chilonga 2021), full TLMA demarcation was important to create trust by traditional authorities who, based on other countries' experience, doubted Government's intention. All TLMAs were demarcated in 2018 to comply with a World Bank Development Policy Loan. This clears the path for clarifying the status of expired estate leases within such areas and subsequent local adjudication and issuance of documented use rights in a low-cost process.

3. Harnessing land policies for broader socio-economic development

Complementary policies can enhance the benefits from land registries and offer independent entry points for reform. Enabling local governments to use property records for property taxation, land value capture, and planning, including for urban expansion can help increase urban job growth. Removing obstacles to rural factor markets functioning, encouraging intergenerational transfers of use rights, and improving the scope for contracting with private parties higher up in the value chain can foster structural change in rural areas. Clarifying rights, including to public land, to foster investment, integration into larger markets, and insurance and publication of land price and climate risk information can strengthen resilience. Each of these areas also offers entry point for reform of land institutions.

3.1 Urban property taxation, service delivery, and expansion

As land is immobile, taxing it allows to raise revenue to finance local public goods in a way that is less distortive (Schwerhoff et al. 2020) and, if the rich hold more of their wealth in land, more progressive (Bonnet *et al.* 2021) than taxes on mobile factors such as capital and labor. Property taxes also increase owners' incentives to use land effectively (Oates and Schwab 1997), discourage speculation (Norregaard 2013), and have reduced vacancies (Segú 2020). They encourage compact urban development (Ermini and Santolini 2017; Song and Zenou 2006) and can be output-enhancing (Kalkuhl and Edenhofer 2017).

Local public goods will increase land values (Coury *et al.* 2021; Gonzalez-Navarro and Quintana-Domeque 2016) and services financed using property tax proceeds will benefit local residents. This implies property taxes strengthen the social contract between state and taxpayers (Besley and Persson 2014). Collection of property tax increased participation and accountability even in fragile low-capacity African settings (Weigel 2020) with willingness to pay such taxes enhanced by better service delivery (Kresch *et al.* 2023). In the longer term, the relative stability of property tax revenue allows it to be used to servicing municipal bonds that, among others, can generate resources to invest in infrastructure.

Country studies including from Ghana (Dzansi *et al.* 2022), DRC (Balan *et al.* 2020), Senegal (Knebelman 2021), and Uganda (Regan and Manwaring 2023) show that property tax collection remains at 10 to 20 percent of potential in most African cities. To address this and better capture what has been referred to as the continent's largest source of untapped revenue (Collier et al. 2017), there is need to establish an urban tax roll with full coverage; put in place market-based valuation that is regularly updated to ensure buoyancy; minimize exemptions (e.g. for untitled or owner-occupied properties) that are difficult to justify as tax-financed local services benefit all residents, and ensure credible enforcement, e.g. by requiring submission of a tax clearance certificate to register any subsequent registration for a given parcel.

Transaction taxes ('stamp duties') have been widely used historically as they allow collecting land revenue from a limited number of urban properties without incurring the cost of maintaining a tax map that is high if paper maps are relied on. The ability to access satellite imagery and mass valuation techniques makes land value capture much easier. Regulation has, however, been slow to adjust and transaction taxes continue to account for the lion's share of mean transfer fees of more than 7 percent across African countries (see table 2). Although the revenue, net of collection cost, they generate is often modest, transfer taxes tend to push transactions into informality so that registries lose currency and foster under-reporting of transaction prices so that reported price no longer convey meaningful information.

Evidence from developed countries shows that even modest transfer taxes are very distortionary: A 1.1% land transfer tax in Toronto is associated with significant welfare losses (Dachis *et al.* 2012) and, if the decision of whether to own or rent is accounted for, estimated to incur a deadweight loss of 79% of revenue (Han *et al.* 2022). In the UK, transaction taxes cause large distortions to the volume, timing, and price of property transactions (Best and Kleven 2018) and negatively affect land markets and labor mobility (Eerola *et al.* 2021; Hilber and Lyytikainen 2017). Replacing them with an equivalent recurrent land tax on a much larger base is easy technically and, especially if provisions such as deferral for high-value properties owned by individuals with low liquid wealth are made, can increase revenue, welfare, efficiency, and transparency.

Replacing high transaction taxes with a recurrent land tax and linking efforts to increase property taxation to visible reductions in registration fees could also remove a major obstacle to establishing and maintaining urban land registries. In settings where regulatory reform as a basis for urban titling is not viable in the short term, establishing a fiscal cadaster can be an attractive option to increase own source revenue, reduce or remove transaction taxes that would be a major obstacle to registry sustainability, generate the data needed to award more secure rights, and allow demand for such rights to be articulated and voiced more effectively.

Satellite imagery has been used to cost-effectively generate urban tax maps at scale in Italy (Casaburi and Troiano 2016) and, via large national programs, in India (Awasthi *et al.* 2020), and Brazil (Christensen and Garfias 2020). Computer-assisted mass appraisal of residential property is standard in most developed countries (Grover et al. 2017). For example, in the Netherlands 2.8% of GDP is raised in property tax revenue by transparently valuing millions of properties every year in a way that is equitable, trusted and perceived as fair as assessed property values are available publicly (Kuiiper and Kaathman 2015). While application of mass valuation techniques to Africa is straightforward (Ali *et al.* 2020b), countries in the region overwhelmingly use formulas (i.e., charges per property or area unit) that are less tightly linked to market value and thus less buoyant and often more regressive. They often require property to be valued by a licensed valuer, a procedure that is more costly and less transparent, fair, and buoyant than mass valuation and may persist only as reported transaction prices cannot be trusted.

A legal or fiscal cadaster is also a precondition for financing infrastructure investment based on changes in land values, a large potential that is only partly utilized in developed (Gupta *et al.* 2022) and developing country contexts (Tsivanidis 2021). Beyond increasing land values, investment in urban transit (Balboni *et al.* 2021) or metros (Du and Zheng 2020; Zárate 2022) generated large social benefits and often expanded access to formal jobs. Such benefits tend to be persistent: labor market access due to proximity to transport links increased inter-generational mobility in the UK (Costas-Fernandez *et al.* 2022).

Beyond urban areas, better roads are critical for market integration as they allow farmers to exit subsistence (Gollin and Rogerson 2014); diversify into perishable high-value crops with higher value added (Pellegrina and Gafaro 2021); and reduce prices for bulky inputs such as fertilizer (McArthur and McCord 2017). Better market access due to railroad expansion in 1870-1890 increased agricultural land values in US counties by an average of 60% (Donaldson and Hornbeck 2016). Trade cost in Africa amount to 5 times the global average and targeted investment to reduce them can yield large welfare gains, including via reduced food prices (Porteous 2019) and promotion of fertilizer use in ways that are more effective and sustainable than subsidies (Porteous 2020). In Indonesia, road maintenance contributed to shifts out of informality, reduced food price fluctuation, and higher land values with benefits three times the cost (Gertler et al. 2022).

In cities, deploying arterial infrastructure ahead of development also allows public services to be provided more cost-effectively, at about one third of what it would cost to provide the same services *ex post* (Smolka and Biderman 2012). As 30–50% of a city's surface is used for public spaces, patterns of public land use and arterial infrastructure are persistent over time (Baruah *et al.* 2021), affect the level of innovation (Roche 2020) and can be used to crowd in private investment: In Tanzania, public provision of arterial infrastructure and basic infrastructure under 'site and service' projects that aimed to combine mixed-income zoning with access to jobs and mobility (Owens *et al.* 2018) in the 1970s helped crowd in private investment by allowing owners to incrementally upgrade structures: 3-4 decades after the program, treated areas had larger buildings, better electricity access, and superior socio-economic outcomes than untreated ones as and slums that had been upgraded at much higher cost (Michaels *et al.* 2021).

The impacts of ill-functioning land registries may be compounded by planning rules make conversion from agricultural to non-agricultural use difficult (Blakeslee *et al.* 2021) or that, by stipulating minimum lot sizes or limits on floor-area ratios (Lall 2017), may impede job creation and urban floorspace investment. In so doing, they increase housing costs and push individuals into informality, imposing a 'labor tax' that reduces labor market participation, especially by women (Field 2007), thus constraining economic opportunities (Cavalcanti *et al.* 2019). As many structures fail to comply with such rules, this creates insecurity and scope for rent seeking (Cai *et al.* 2017) while making it more difficult to implement needed planning regulations that could help make cities more resilient to climatic shocks.

High cost of urban floorspace encourages informal settlement formation on private (Brueckner and Selod 2009; Brueckner 2013) or public (Shah 2014) land at cities' fringe with poor access to services, often in hazardous locations. Slums may be organized by politically (Obala and Mattingly 2014) or ethnically (Marx *et al.* 2019) connected individuals in a way that may result in lock-in effects and reduces welfare of those newly arriving in cities: In Kibera, Nairobi's main slum, a model of the housing markets estimates benefits from redevelopment to be more than 30 times the typical annual slum rent payment by all slum households even after fully compensating landlords (Henderson *et al.* 2021). Redevelopment of informal areas on about 15% of Mumbai's area in 2006-10 resulted in large but unevenly distributed gains (Gechter and Tsivanidis

2020), implying a need to compensate those losing out, e.g. through transferable vouchers (Kumar 2021) or mixed income housing as part of the redevelopment (Blanco and Neri 2022).

A combination of tenure regularization and better service provision is often suggested to improve conditions by current residents. Such a policy needs to be carefully designed to avoid inviting land invasions expecting future regularization. Also, restricting the land rights provided to slum dwellers can inadvertently block future development: for example, thirty years after a large scale slum upgrading program that affected more than 5 million Indonesian slum dwellers in 1969-84, project areas were more informal and denser with 50% fewer high-rises; more fragmented land holdings; and lower land values than untreated neighboring areas, a finding attributed to the fact that land rights awarded to program beneficiaries were non-transferable (Harari and Wong 2019). Experiments in Latin America that provided randomly selected poor slumdwellers with better housing without property rights similarly did not directly affect asset ownership or labor market outcomes (Galiani *et al.* 2017) and had only transitory indirect effects (Galiani *et al.* 2021).

3.2 Improved rural factor market functioning

In rural areas, immobility of land, informational imperfections (Britos *et al.* 2022) and enforcement cost, create land market frictions. This is illustrated in Kenya where a RCT that subsidized short-term rentals increased rental by younger, land-poor, and more entrepreneurial farmers, resulting in higher output and value added on rented plots via more intensive application of non-labor inputs (Acampora *et al.* 2022). Imperfections in land and other factor markets, in a non-mechanized environment most notably those for labor rooted in the difficulty of observing effort caused by seasonal nature and spatial dispersion of agricultural activities, have long been shown to lead to a negative relationship between land size and productivity (Feder 1985).

Owner-operators' residual claims to profit can provide them with a supervision cost advantage over wage-labor operated farms (Binswanger *et al.* 1995). Alternatively, if farm size is too small to fully employ owners' family, frictions associated with participation in off-farm labor markets may lead to seasonal under-employment (Dillon *et al.* 2019) and prompt farmers to work on their farm even at returns below the market wage, as documented in Rwanda (Ali and Deininger 2015), Malawi, Tanzania, and Uganda (Julien *et al.* 2019). In India, consistent with barriers to participation in off-farm labor markets, non-separability affects small but not large farms (Merfeld 2020) with estimated shadow wages for family labor 20% below non-agricultural wages (Caunedo and Kala 2021).

Even if measurement error can account for part of it (Gollin and Udry 2020), resource misallocation in agriculture is much larger than elsewhere in the economy (Adamopoulos and Restuccia 2014), with land and factor market imperfections a key reason (Restuccia and Rogerson 2017). In Ethiopia where policies prohibiting land sales and restricting leasing in some regions increase market frictions with a hypothetical scenario of unrestricted marketability of agricultural land is estimated to increase GDP by 9% and decrease agricultural employment by 20% (Gottlieb and Grobovsek 2019) in a way that is pro-poor (Restuccia 2021) and could increase zone-level agricultural productivity by 43% (Chen *et al.* 2017).[18]

Incomplete markets for credit and insurance may, especially with more frequent climatic shocks, prompt farmers to minimize risk in ways that may reduce output in the short term and, via liquidation of productive assets or under-investment in human capital (e.g. by pulling children out of school) undermine their coping ability in the long term (Aragon *et al.* 2021). Fragmenting holdings to exploit micro-variation in climate in ecologically varied settings such as Rwanda (Ali *et al.* 2019b) or Ethiopia (Knippenberg *et al.* 2020) allowed farmers to diversify risk, albeit at the cost of having to spend more time traveling between parcels.

Despite efficiency losses as documented for Malawi (Ricker-Gilbert *et al.* 2019) and elsewhere in Africa (Kalkuhl *et al.* 2020), sharecropping can be a second-best risk-sharing mechanism (Otsuka *et al.* 1992) for farmers who are liquidity constrained and cannot access other options for insurance. This is also illustrated in Uganda where randomly assigned higher output shares incentivized tenants to use more inputs and cultivate riskier crops, resulting in a 60% output gain (Burchardi *et al.* 2019) and in Bangladesh where improving credit access and lowering their risk exposure prompted tenants' to choose fixed rent over share contracts (Das *et al.* 2019). Bundling cash rent contracts with insurance could potentially offer large gains.

As the price of capital relative to labor declines, indivisibility of capital inputs that can also help reduce supervision cost (Deininger and Byerlee 2012) may counter family farmers' labor cost advantages and lead to a positive relationship between farm size and productivity (Eswaran and Kotwal 1986), in line with cross-country patterns and historical evidence from the US (Chen 2020). A combination of imperfections in labor and capital markets can give rise to a U-shaped relationship between farm size and productivity as indeed confirmed in India. In this setting, eliminating land market frictions to allow parcel sizes large enough to use machinery could increase income per agricultural worker by 68% and reduce the number of farms by 82% (Foster and Rosenzweig 2021). Better factor market functioning also weakened or reversed the negative farm size-productivity relationship in China (Sheng *et al.* 2019), Bangladesh (Gautam and Ahmed 2019), India (Deininger *et al.* 2018), Vietnam (Liu *et al.* 2020), Brazil (Helfand and Taylor 2021), Tanzania (Wineman and Jayne 2021), Kenya (Muyanga and Jayne 2019), and Nigeria (Omotilewa *et al.* 2021).

The scope for land policies to improve rural factor market functioning in the long term is illustrated by the large impact of China's reforms (Adamopoulos *et al.* 2022): Following the large gains from issuing gradually longer use rights from 1978 (Lin 1992) reducing the scope for periodic land reallocation that had

[18] Globally, such a scenario (via 'titling') is estimated to potentially increase output by as much as 82.5% (Chen 2017).

reduced migration incentives (Giles and Mu 2018) increased off-farm labor supply and rural incomes (Zhao 2020). Reforms to facilitate land rental fostered structural transformation (Deininger and Jin 2009), triggering estimated increases of output and productivity by 8% and 10%, respectively (Chari *et al.* 2021). Complete land registration on a pilot basis almost doubled the likelihood of non-farm enterprise startups (Deininger *et al.* 2019). Better factor market functioning allowed adjustments to offset some of the negative impacts of climate change (Chen and Gong 2021) while cities with more elastic housing supply were able to attract more skilled migrants and industrialize (Niu *et al.* 2021).

Historically, land reforms in East Asia contributed to off-farm development and migration (Kitamura 2022) via credit access (Iscan 2018) and capital accumulation (Hayashi and Prescott 2008) based on secure and fully transferable rights (Jeon and Kim 2000). Land reforms elsewhere were often undertaken in response to political events (Bhattacharya *et al.* 2019) If political objectives received priority (Caprettini *et al.* 2021; Fergusson 2013), politicized beneficiary selection (Fergusson *et al.* 2022) and restrictions on land rights or markets to make reform viable (Adamopoulos and Restuccia 2020) or ensure beneficiary support post-reform (Albertus *et al.* 2016; Albertus 2019) often compromised economic impact. Incomplete (Deininger *et al.* 2004) or overlapping land rights (Deininger and Ali 2008) may impede investment and land markets (Bird and Venables 2020) while bureaucratic implementation reduced size and impact of land reforms implemented in South Africa after the end of apartheid (Keswell and Carter 2014).

Data for the 1870-1910 period for the US point to generational change as driver of structural transformation (Porzio *et al.* 2022), highlighting the importance of reducing barriers to intergenerational land transfers that are reported from Ethiopia (Kosec *et al.* 2018) or Nigeria (Amare *et al.* 2021) so young farmers can establish viable farms early (Ricker-Gilbert and Chamberlin 2018) and enter into long-term contracts for land to invest (Fan and Yeh 2019).

The surge in of demand for large scale land-based investment (LSLBI) in the wake of the 2007/08 spike in food and commodity prices was expected to provide land abundant African countries with opportunities to access know-how and markets (Collier and Venables 2012). In similar circumstances, establishment of oil palm after 2000, though at high cost (Li and Semedi 2021), lifted more than a quarter of Indonesia's rural individuals out of poverty (Edwards 2019a); increased the productivity of manufacturing firms well beyond the oil palm value chain (Kraus *et al.* 2021); and improved access to employment, infrastructure (Edwards 2019b), nutrition, living standards and human capital formation (Chrisendo *et al.* 2022).

Although cultivated area in Africa increased by some 50 million ha between 2013 and 2019, well above the increase in Asia, studies point to limited direct or indirect benefits: Large farm establishment in Ethiopia did not create jobs and had little effect on smallholders' access to technology or input and insurance markets (Ali *et al.* 2019a). In Mozambique, use of improved practices and inputs by smallholders adjacent to large farms increased but yields or market participation did not (Deininger and Xia 2016) and large farms may put downward pressure on wages (Glover and Jones 2019). In Zambia, spillovers from large farm establishment were at best limited and competition for land increased (Lay et al. 2020).[19]

The literature suggests that, in addition to non-traditional investors focusing on countries with weak land governance (Arezki *et al.* 2015), limited impacts of LSLBI can partly be attributed to the use of centralized and non-competitive mechanisms for investor identification and land transfer that often prioritized political (Bélair 2021) or territorial (Lavers 2016) over economic objectives.[20] Land centrally assigned to investors was often found occupied ('encroached'), mired in conflict, or less suitable than expected. Respect for existing rights (Maganga *et al.* 2016), safeguards (Nolte and Voget-Kleschin 2014), or disclosure (Cotula 2014) and job creation or skill transfer (Nanhthavong *et al.* 2022) often suffered. While delays and bureaucratic bickering led some investors to withdraw (Engström and Hajdu 2018), those that proceeded were often unable to mitigate resulting environmental issues (Shete *et al.* 2016) or conflict (Sulle 2020).

Non-competitive ways of land allocation increased the likelihood of selecting investors with limited skills and made it difficult for productive land users, including local farmers, to acquire land from unsuccessful investors via market-based processes, especially if unused land was expected to revert to central agencies. In Zambia, (titled) large farms participated less in land markets than informal small ones and failed to access credit or achieve higher productivity. Beyond speculative land acquisition often by politically connected individuals (Sitko and Jayne 2014), high cost of registering transfers (including survey cost) likely also plays a role (Ali and Deininger 2022b). Enforcing payment of ground rent including to legally extinguish dormant titles by absentee landlords based on non-payment is important to ensure that expansion of medium-scale farms' (Jayne et al. 2019) will increase productivity and local welfare. Given smallholders' advantage in labor-intensive products with high value-added (Rogers *et al.* 2021), emphasis on upstream investment that has shown to generate larger income or poverty gains than investment focused on land only (Van den Broeck *et al.* 2017) will offer greater scope for market integration of smallholders (Dubbert (2019) and a more integrated investment approach (Schoneveld 2022).

While renewable energy is becoming increasingly competitive with traditional sources (Chakravarty and Somanathan 2021), it may be land intensive. If land rights are not clearly defined, land demand for green energy or payments for environmental services (PES), can threaten to dispossess current users (Kansanga and Luginaah 2019). If the size of incentives exceeds the opportunity cost of environmentally inferior uses and collective action (Edwards *et al.* 2021) or political economy (Cisneros *et al.* 2021; Pailler 2018) issues can be resolved, competitively awarded PES can possibly incentivize market-driven provision of

[19] Beyond Africa, similar results are reported from Cambodia (Anti 2021), and the Lao People's Democratic Republic (Nanhthavong *et al.* 2020).
[20] Case studies show that political objectives may change quickly (Widengard 2019) or not be shared by local officials (Dieterle 2021).

environmental services in socially desirable ways. However, such Evaluation of PES schemes (Börner *et al.* 2017) suggests performance can be enhanced by a stronger focus on sustainability; additionality, i.e., PES target areas that, without support, are at high risk of irreversible land use change; and avoidance of leakage, i.e., ensuring higher levels of conservation in target areas is not offset by lower ones in non-target areas; and effects are sustained over time (Chabe-Ferret and Voia 2021). Allocating PES subsidies via competitive auction can help reveal private information including on strategic complementarities between neighboring parcels (Lewis and Polasky 2018). Auctions were used effectively in developing country settings such as Malawi (Ajayi *et al.* 2012) where they reduced conservation cost by about 30% (Jack 2013). PES achieved temporary reductions in deforestation in Uganda (Jayachandran *et al.* 2016) that are economically viable at a cost of carbon of $39/t.

3.3 Resilience to climate change

Its spatial extension makes agriculture more vulnerable to climate-induced temperature extremes or weather shocks than industry or services. Globally, changes in climate are estimated to have reduced TFP in agriculture by 21 percent, equal to 7 years of productivity growth, with effects concentrated in warmer regions (Ortiz-Bobea et al. 2021). Extreme heat waves may cause significantly larger damage than changes in means (Miller et al. 2021). Spatially explicit two-sector models with costly trade, innovation, technology diffusion, and land values that capture amenities and external effects from human capital or carbon stocks have thus been long used to analyze global impacts under different emission pathways (Cruz and Rossi-Hansberg 2021; Desmet and Rossi-Hansberg 2015) that point to structural transformation, trade, and migration as important strategies for adaptation. Incorporating an explicit production possibility frontier (Costinot *et al.* 2016) points to crop choice as another adaptation strategy. With subsistence constraints climate change may pull labor into rather than out of the agricultural sector with low productivity (Nath 2020). Conversely, gains from lower trade costs could possibly offset predicted losses from climate change (Porteous 2023) and equalization of intra-Africa's trade and migration cost with those in the EU may even reverse them (Conte 2022).

The scope to diversify risks via integration into market is illustrated by analysis of drought impacts on African cities over five decades: those with firms selling output to global markets were able to absorb shocks and even expand, while those dependent on local markets only suffered declines as demand dried up (Henderson *et al.* 2017). Farmers can adjust to temporary shocks through land rental (Eskander and Barbier 2023) or precautionary savings as in India where rural bank branch expansion mitigated drops in rural productivity and life expectancy due to an increased number of hot days (Burgess et al. 2017).

Persistent shocks seem, however, less amenable to intra-sectoral adjustment: in one Indian state, persistent drying up of wells reduced farm income but these losses could, especially in areas with well-developed manufacturing sectors, be fully compensated through. However, households had to reduce asset holdings, making them more vulnerable in the longer term (Blakeslee *et al.* 2020).[21] Over six decades, census data show higher temperatures decreased agricultural productivity, reducing demand for nonagricultural goods or services, and triggering out-migration (Liu *et al.* 2023). In fact, illiquid land markets and cultural norms seem to have left landowners with a 'stranded asset', triggering behavior quite distinct from that of non-landowners (Basu 2022). In Brazil, integration with other regions provided insurance for short-term climatic shocks but long-term climate-induced productivity declines triggered capital outflows and outmigration as local manufacturing could not absorb all displaced workers and the importance of social networks points towards labor market frictions (Albert *et al.* 2021).

Resilience can be enhanced and the need for defensive spending by farmers (Jagnani *et al.* 2021) reduced via research and investment in irrigation or mechanization that will be increased by active land and credit markets, a plausible pathway for the observed impact of institutional reforms on observed crop 'yield gaps' at country borders obtained from remotely sensed imagery (Wuepper *et al.* 2023). However, knowledge of appropriate technology for on-farm water management may also constrain farmers' behavior as in Niger where training on water conservation increased output by some 80%, triggering spillovers to neighbors (Aker and Jack 2021). Building on land rights, institutional innovation to improve use of existing water resources can also have large impact as in the US where a establishing tradable water rights prompted increased water use efficiency and shifts to high-value crops with benefits more than four times the cost (Browne and Ji 2023).

At the city level, extreme weather events such as wildfires (Baylis and Boomhower 2021) or hurricanes (Bakkensen and Blair 2022) can create externalities that can be addressed via building codes or other form of regulation. As many cities are situated in floodplains, climate change will affect urban land use and values via higher frequency of extreme weather events such as cyclones or floods, changes in amenity values due to higher temperatures, and long-term susceptibility to sea level rise. In the absence of adaptation investment or potential for migration, sea level rise may displace not only displace people but also reduce GDP significantly (Desmet et al. 2021).

Extreme weather events affect property prices and land values in the US (Bernstein *et al.* 2019) and the UK (Beltran *et al.* 2019), though effects by individuals may be temporary (Keys and Mulder 2020) and affected by awareness (Baldauf *et al.* 2020) or beliefs (Nguyen et al. 2022). Insurers, mortgages lenders, and institutional investors are better at pricing risk appropriately (Atreya and Czajkowski 2019) and will have broader effects if, for example, insurance is mandatory in certain high risk areas and property prices are public. Prices for municipal bonds have started to reflect the risk of sea level rise over and above specific

[21] Other studies support the importance of access to non-agricultural income, through labor reallocation (Colmer 2021), public employment (Fetzer 2020), or inter-state migration (Sedova and Kalkuhl 2020).

property prices (Goldsmith-Pinkham et al. 2023). Yet, institutional and political economy factors imply that, even in the US, publicly available information on risk is not fully reflected in property prices (Hino and Burke 2021). Such informational gaps may result in some properties becoming uninsurable (Bakkensen and Ma 2020) or create political pressure for even more costly and regressive ex-post disaster support (Billings *et al.* 2022). Ensuring that localized risk is properly communicated, e.g., via flood risk disclosure laws (Gourevitch et al. 2023) will thus be important to encourage private adaptation (Barrage and Furst 2019), in line with evidence on publishing data on other types of hazards (Greenstone et al. 2022).

In African cities, higher frequency of floods already causes costly travel delays (He et al. 2021). Yet, though it is not difficult to identify sites with high levels of flood risk (Mulder 2022), the housing stock susceptible to such risk keeps expanding with low- and middle-income countries making up more than 80% of the increase (Rentschler et al. 2022). Better data to formulate land use restrictions and real-time enforcement using earth observation can help prevent construction in risky areas (Boustan *et al.* 2012), often public land.

Subsidies to flood insurance are likely to encourage moral hazard - and have been shown to transfer wealth to the rich in the UK (Garbarino *et al.* 2023). This is similar to moral hazard and possibly large implicit subsidies due to public wildfire protection for residential homes (Baylis and Boomhower 2023) or crop insurance in agriculture (Annan and Schlenker 2015). The case of Jakarta's seawall illustrates that, unless government can credibly commit, moral hazard that crowds out private adaptation efforts can result from any public adaptation investment (Hsiao 2021). The ability to use flexible fiscal instruments such as property tax surcharges levied in proportion to the benefits accruing to individual land parcels from such investment can make it easier for governments to commit and, if used consistently in evaluating and debating large investments and their implications for local property values *ex ante*, also increase the quality of investment decisions and the legitimacy and credibility of such commitment.

4. Conclusion and policy implications

While many studies analyze the impact of issuing titles to private land, the regulations governing operation of the registries that manage, enforce, and publicize such rights will affect the long-term benefits from such interventions as well as private demand for unsubsidized titles. Such registries also support economic development more broadly by providing information on land values and use needed for land taxation or value capture to support local service provision or finance infrastructure projects, allow land for such projects to be acquired to expeditiously, and regulate and plan land use, and prevent abuse of public land.

Improving registry performance is made easier by innovations in technology that allow to reduce registry cost and improve data quality via interoperability; monitor land use in near real time via earth observation; and interact with landowners or -users using mobile connectivity. Fully harnessing this potential will require changes in the regulatory framework to, among others (i) increase publicity of information on land rights and prices to support liquid land, financial, and insurance markets; (ii) make information on location-specific risks widely available and require disclosure so it can be reflected in market prices, insured against, and used by private actors or local governments to adjust and adapt land use in cost-minimizing ways; (iii) allow the use of earth observation for public land monitoring and enforcement of land use regulations; and (iv) enable land owners or users to use digital IDs and mobile connectivity to update or transact rights, share data on their land rights and contract with to third parties (e.g. banks, input suppliers, or processors of output they produce). The importance of regulations to support effective registry operation as an enabler for private sector activity suggests it may warrant greater attention in country strategies a key element of state capacity.

The potential for reforms to improve functioning of markets for land, credit, and insurance via affordable survey standards, digital data integration and workflow management, lower transfer taxes, and low-cost ways of mortgage registration and enforcement, is highest in urban areas. If regulations to allow efficient registry operation are in place, such registries can be established at low cost and sustained on a cost recovery basis. If a complete overhaul of survey standards and procedures for registration is too complex or might face resistance, adjusting valuation standards and creating a complete digital tax roll can be a first step that can be accomplished more quickly and at lower cost. Especially if combined with computerized mass valuation based on market values, minimizing exemptions, and ensuring consistent enforcement of property tax payment, this can allow to generate tangible resources quickly and result in stronger demand for secure rights as one part of the social contract.

In rural areas, regulating decentralized issuance of use rights is, in most African countries, possible under existing laws and can quickly expand coverage with georeferenced use rights to improve farmers' ability to mitigate risk, access markets and working capital, and enter long-term intergenerational land transfers to support structural transformation. Integrating data fully with those from 'farmer registries' or other data bases that are now widely used to administer agricultural support that are often poorly maintained or lack a spatial reference can help address misuse and generate synergies by targeting subsidies better and focusing them on improving and complementing market functioning and rather than distorting them, possibly with negative impacts on environmental sustai8nability and governance. Local and customary authorities can play an important role not only in managing use rights but also to help better utilize common land and resolve land conflict if their boundaries are clearly demarcated; their access to information is improved and their capacity to use technology strengthened; if they have incentives to do so, e.g., by being able to retain some property tax revenue; if mechanisms to ensure transparency and accountability and prevent abuse of authority and conflict of interest are in place.

Evidence from Africa and beyond highlights that land institutions affect climate resilience, urbanization, structural transformation, female empowerment, and governance. This cross-cutting nature adds to their technical complexity and, by expanding the range of stakeholders who need to be consulted, makes it easier for specific groups to oppose or delay reform. A set of jurisdiction-level regulatory indicators akin to the data historically collected by the World Bank's 'Doing Business' project,[22] complemented by administrative data to measure outcomes (e.g., number of registered rights, transactions, mortgages, and land-related court cases) in a way that is more actionable, can help motivate reforms and track progress with implementation in a way that can serve as a point of reference for country strategy documents.

[22] The 'Registering Property' indicator under the World Bank's 'Doing Business' Project collected information on indicators of registry quality as part of a 'Land Administration Quality Index' since 2015. While few large policy changes and limited granularity impliy the index changed little over time, it has been linked to economic growth (d'Arcy, Nistotskaya and Olsson 2021) and urban form (Djankov *et al.* 2021) in a cross-section.

Table 1: Access to title for residential and agricultural land in selected countries (share of respondents)

	Residential land								Agricultural land			
	Own	**Title**	**Updated**	**Male**	**Q1**	**Q2**	**Q3**	**Q4**	**Own**	**Rent**	**Free**	**Title**
BEN	0.536	0.324	0.270	0.751	0.163	0.256	0.358	0.589	0.684	0.103	0.295	0.012
BFA	0.740	0.271	0.659	0.866	0.153	0.231	0.244	0.582	0.776	0.025	0.361	0.024
CIV	0.430	0.361	0.529	0.808	0.265	0.327	0.400	0.524	0.860	0.081	0.155	0.076
GNB	0.720	0.327	0.526	0.722	0.210	0.28	0.359	0.508	0.911	0.037	0.215	0.043
MLI	0.794	0.327	0.517	0.880	0.138	0.243	0.409	0.667	0.899	0.022	0.118	0.010
NER	0.813	0.111	0.627	0.840	0.031	0.064	0.122	0.343	0.843	0.052	0.219	0.015
SEN	0.717	0.557	0.533	0.783	0.327	0.492	0.620	0.805	0.813	0.045	0.302	0.018
TCD	0.830	0.123	0.438	0.714	0.061	0.115	0.142	0.247				0.012
TGO	0.432	0.248	0.633	0.716	0.096	0.151	0.257	0.529	0.577	0.204	0.341	0.013

Source: Calculations using data from West African Economic and Monetary Union surveys.
Note: Countries included are Benin (BEN), Burkina Faso (BFA), Côte d'Ivoire (CIV), Guinea Bissau (GNB), Mali (MLI), Niger (NER), Senegal (SEN), Tchad (TCD) and Togo (TGO). Q1-Q4 refers to the household's quartile of per capita expenditure. For agricultural land, own, rent, and free denote the modality of land access.

Table 2: Key attributes of land administration systems in African countries

Region	Registration cost (%)	Digital records			Main city covered		Court dec. in < 1 a	Number of countries
		Registry	Cadaster	Linked	Registry	Cadaster		
Africa, Central	9.3	8.3	8.3	8.3	8.3	8.3	8.3	12
Africa East	5.3	14.3	14.3	14.3	0.0	0.0	14.3	7
Africa, North	4.7	0.0	16.7	0.0	0.0	16.7	0.0	6
Africa, South	6.9	8.3	0.0	0.0	33.3	25.0	8.3	12
Africa, West	7.2	0.0	6.3	0.0	6.3	6.3	6.3	16
SSA	7.02	5.7	7.5	3.8	11.3	11.3	7.5	53
EAP	4.5	20.8	33.3	20.8	58.3	54.2	29.2	24
ECA	2.7	43.5	52.2	30.4	60.9	69.6	39.1	23
WENA	4.2	58.8	88.2	55.9	91.2	97.1	29.4	34
LAC	5.9	3.1	31.3	3.1	15.6	31.3	3.1	32
ME	5.8	28.6	28.6	14.3	71.4	64.3	7.1	14
SAS	7.0	12.5	25.0	12.5	25.0	25.0	12.5	8
World	5.4	23.4	37.2	19.7	43.6	47.3	17.6	188

Source: Calculations using data from 2020 Registering Property data.

Note: Figures in column 2 to 7 refer to the percentage of countries that score on the respective indicator (e.g., 8.3 in line 1 column 2 means that 8.3% of the 12 countries in Central Africa (i.e., one country) has a digital registry. Court decision in < 1 a means that a first instance court decision for the most common type of land dispute can be obtained in less than one year. EAP = East Asia & Pacific; ECA = Eastern Europe & Central Asia; LAC = Latin America & Caribbean; ME = Middle East; SAS = South Asia; SSA = Sub-Saharan Africa; WENA = Western Europe and North America.

Table 3: Perceptions of reliability and transparency of land administration in African regions (share of respondents)

Region	Can get land information	Fraudulent registration possible by Ordinary people	Rich people	Number of countries
Africa, Central	0.197	0.158	0.754	3
Africa, East	0.208	0.120	0.656	3
Africa, North	0.093	0.167	0.538	3
Africa, South	0.210	0.194	0.603	10
Africa, West	0.189	0.161	0.598	15
Total	0.189	0.167	0.613	34

Source: Afrobarometer Round 7.

Note: Can get land information is the share responding "very likely" to the question "If you went to the local lands office to find out who owns a piece of land in your community, how likely is it that you could get this information?" *Fraudulent registration (ordinary and rich)* tabulates the share of respondents who think it is very likely for an ordinary or a rich person, respectively, to "…pay a bribe or use personal connections to get away with registering land that does not belong to them." *Women's rights* is the share who "agree or strongly agree" with the statement "Women should have the same rights as men to own and inherit land."

Table 4: Actual tasks and desired involvement by traditional leaders (share of respondents)

Region In Africa	Traditional leader's influence should		Traditional leaders play a main role in				No. of countries
	Increase	Increase a lot	Dispute resolution	Land allocation	Governing community	Influencing people's vote	
Center	0.653	0.265	0.319	0.232	0.196	0.181	2
East	0.514	0.272	0.488	0.249	0.297	0.158	3
North	0.518	0.293	0.316	0.156	0.272	0.196	1
South	0.556	0.231	0.521	0.310	0.453	0.131	5
West	0.543	0.250	0.522	0.359	0.366	0.211	12
Total	0.551	0.252	0.490	0.314	0.357	0.183	23

Source: Own computation from Afrobarometer Round 8 data.
Note: Figures in columns 3 to 6 are the share of respondents indicating that traditional leaders have "a lot" of influence over dispute resolution, land allocation, governing the community, or influencing people's vote, respectively.

Table 5: Share of violent conflicts over land and natural resources

	No. of violent events		
Region	**2007–12**	**2013–17**	**2018–22**
Africa, Central	0.021	0.049	0.083
Africa, East	0.047	0.060	0.072
Africa, North	0.022	0.041	0.047
Africa, South	0.029	0.037	0.054
Africa, West	0.027	0.052	0.098
All Africa	0.032	0.048	0.073

Source: Calculations based on data from the Armed Conflict Location & Event Data Project.
Armed Conflict Location & Event Data Project (ACLED) data (Raleigh et al. 2010) are available at www.acleddata.com, for a total of 281,311 violent conflicts (with at least one fatality) in Africa between January 1, 1997, and May 27, 2022. A conflict event is classified as land related if the event description includes "land" but not "landmines" or "landed" and as resource related if the description includes "farm," "crop," "harvest," "pastoral," "livestock," "grazing," "pasture," "cattle," or "cow." The share is used to account for the fact that, for example due to better access to social media, the number of violent conflicts reported in the ACLED data increased dramatically between 2007–12 and 2017–22.

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